

Connect and Protect: Networks and Network Security

Module 1

Network architecture

Get started with the course

Introduction to Course 3

Video • 1 min

Course 3 overview

Reading • 4 min

Helpful resources and tips

Reading • 4 min

Introduction to networks

Welcome to module 1

Video • 29 sec

Chris: My path to cybersecurity

Video • 3 min

What are networks?

Video • 1 min

Tina: Working in network security

Video • 2 min

Emmanuel: Useful skills for network security

Video • 2 min

Network tools

Video • 2 min

Network components, devices, and diagrams

Reading • 4 min

Practice: Design a local area network

Ungraded Plugin • 10 min

Cloud networks

Video • 2 min

Cloud computing and software-defined networks

Reading • 8 min

Identify: Functions of network tools

Ungraded Plugin • 10 min

Test your knowledge: Introduction to networks

Practice Assignment • Grade: 100%

Network communication

Introduction to network communication

Video • 2 min

The TCP/IP model

Video • 2 min

The four layers of the TCP/IP model

Video • 1 min

Learn more about the TCP/IP model

Reading • 4 min

The OSI model

Reading • 8 min

Test your knowledge: Network communication

Practice Assignment • Grade: 100%

Local and wide network communication

IP addresses and network communication

Video • 2 min

Components of network layer communication

Reading • 4 min

Test your knowledge: Local and wide network communication

Practice Assignment • Grade: 100%

Review: Network architecture

Wrap-up

Video • 39 sec

Glossary terms from module 1

Reading • 4 min

Module challenge 1

Graded Assignment • Grade: 100%

Module 2

Network operations

Introduction to network protocols

Welcome to module 2

Video • 1 min

Network protocols

Video • 3 min

Common network protocols

Reading • 4 min

Additional network protocols

Reading • 4 min

Arturo: Working in network security

Video • 2 min

Wireless protocols

Video • 1 min

The evolution of wireless security protocols

Reading • 4 min

Test your knowledge: Recognize different network protocols

Practice Assignment • Grade: 100%

System identification

Firewalls and network security measures

Video • 4 min

Virtual private networks (VPNs)

Video • 2 min

Security zones

Video • 3 min

Subnetting and CIDR

Reading • 4 min

Proxy servers

Video • 3 min

Virtual networks and privacy

Reading • 4 min

VPN protocols: Wireguard and IPSec

Reading • 4 min

Test your knowledge: System identification

Practice Assignment • 10 min

Review: Network operations

Wrap-up

Video • 51 sec

Glossary terms from module 2

Reading • 4 min

Module challenge 2

Graded Assignment • 50 min

Module 3

Secure against network intrusions

Module 4

Security hardening

VPN protocols: Wireguard and IPSec

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A VPN, or virtual private network, is a network security service that changes your public IP address and hides your virtual location so that you can keep your data private when you're using a public network like the internet. VPNs provide a server that acts as a gateway between a computer and the internet. This server creates a path similar to a virtual tunnel that hides the computer's IP address and encrypts the data in transit to the internet. The main purpose of a VPN is to create a secure connection between a computer and a network. Additionally, a VPN allows trusted connections to be established on non-trusted networks. VPN protocols determine how the secure network tunnel is formed. Different VPN providers provide different VPN protocols.

This reading will cover the differences between remote access and site-to-site VPNs, and two VPN protocols: WireGuard VPN and IPSec VPN. A VPN protocol is similar to a network protocol: it's a set of rules or instructions that will determine how data moves between endpoints. An endpoint is any device connected on a network. Some examples of endpoints include computers, mobile devices, and servers.

Remote access and site-to-site VPNs

Individual users use remote access VPNs to establish a connection between a personal device and a VPN server. Remote access VPNs encrypt data sent or received through a personal device. The connection between the user and the remote access VPN is established through the internet.

Enterprises use site-to-site VPNs largely to extend their network to other networks and locations. This is particularly useful for organizations that have many offices across the globe. IPSec is commonly used in site-to-site VPNs to create an encrypted tunnel between the primary network and the remote network. One disadvantage of site-to-site VPNs is how complex they can be to configure and manage compared to remote VPNs.

WireGuard VPN vs. IPSec VPN

WireGuard and IPSec are two different VPN protocols used to encrypt traffic over a secure network tunnel. The majority of VPN providers offer a variety of options for VPN protocols, such as WireGuard or IPSec. Ultimately, choosing between IPSec and WireGuard depends on many factors, including connection speeds, compatibility with existing network infrastructure, and business or individual needs.

WireGuard VPN

WireGuard is a high-speed VPN protocol, with advanced encryption, to protect users when they are accessing the internet. It's designed to be simple to set up and maintain. WireGuard can be used for both site-to-site connection and client-server connections. WireGuard is relatively newer than IPSec, and is used by many people due to the fact that its download speed is enhanced by using fewer lines of code. WireGuard is also open source, which makes it easier for users to deploy and debug. This protocol is useful for processes that require faster download speeds, such as streaming video content or downloading large files.

IPSec VPN

IPSec is another VPN protocol that may be used to set up VPNs. Most VPN providers use IPSec to encrypt and authenticate data packets in order to establish secure, encrypted connections. Since IPSec is one of the earlier VPN protocols, many operating systems support IPSec from VPN providers.

Although IPSec and WireGuard are both VPN protocols, IPSec is older and more complex than WireGuard. Some clients may prefer IPSec due to its longer history of use, extensive security testing, and widespread adoption. However, others may prefer WireGuard because of its potential for better performance and simpler configuration.

Key Takeaways

A VPN protocol is similar to a network protocol: it's a set of rules or instructions that will determine how data moves between endpoints. There are two types of VPNs: remote access and site-to-site. Remote access VPNs establish a connection between a personal device and a VPN server and encrypt or decrypt data exchanged with a personal device. Enterprises use site-to-site VPNs largely to extend their network to different locations and networks. IPSec can be used to create site-to-site connections and WireGuard can be used for both site-to-site and remote access connections.

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Venkatesh

Give me a summary

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This reading explains what a Virtual Private Network (VPN) is, how it works, and differentiates between various types and protocols.

Understanding VPNs

- A VPN is a network security service that masks your public IP address and virtual location, ensuring data privacy on public networks.
- It functions by creating a secure, encrypted tunnel between your device and the internet via a VPN server.

Types of VPNs

- Remote Access VPNs:** Used by individuals to secure connections between personal devices and a VPN server, encrypting data sent or received.
- Site-to-Site VPNs:** Employed by enterprises to extend their network securely to multiple locations, often using IPSec for encrypted tunnels.

VPN Protocols: WireGuard vs. IPSec

- WireGuard VPN:** A newer, high-speed, and open-source protocol known for its advanced encryption, simpler setup, and faster download speeds.
- IPSec VPN:** An older, more complex protocol widely supported by operating systems, valued for its extensive security testing and widespread adoption.

And if you want to continue exploring this topic, try one of these follow-up questions:

What is the difference between remote access and site-to-site VPNs?

How can you set up a WireGuard VPN for faster internet access?

What are the advantages and disadvantages of using IPSec versus WireGuard VPN?

Ask me anything

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